

PROJECT SUMMARY

Hotel Booking, Revenue and Cancellation Analysis

Team	Analytics hub
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1. Business Problem

The project investigates hotel booking behaviour, with a focus on booking volume, cancellations, customer types, ADR, booking lead time and country-level booking patterns. High cancellation levels can make room availability, staffing, revenue, pricing and hotel operations difficult to plan. The goal is to identify cancellation risk and support actions that reduce cancellations without significantly reducing bookings.

2. Dataset

The project uses the Hotel Booking Demand Dataset. The project workflow uses the cleaned hotel booking data containing hotel, arrival, guest, customer, lead-time, ADR, stay and reservation-status information.

3. Data Cleaning and Transformation

- Inspected dataset dimensions, columns, data types, missing values, duplicates and descriptive statistics.
- Handled missing values where required and checked unusual or invalid values.
- Converted relevant date fields and corrected data types where required.
- Created Total_Guests = Adults + Children + Babies.
- Created Stay_Nights = Weekend Nights + Week Nights.
- Created Booking_Status with Cancelled and Completed categories.

4. Key Findings

- **Booking volume:** City Hotel is the highest booking hotel according to the project analysis.
- **Cancellation level:** The dataset contains 87,389 total bookings, with 63,364 successful bookings and 24,025 cancelled bookings.
- **Hotel cancellation:** City Hotel has the highest cancellation level among the two hotels.
- **Customer value:** Transient customers have the highest average ADR.
- **Monthly booking volume:** August has the highest number of bookings.

KPI	Result
Total Bookings	87,389
Successful Bookings	63,364
Cancelled Bookings	24,025
Highest Booking Hotel	City Hotel
Highest Cancellation Hotel	City Hotel
Highest Average ADR Customer Type	Transient
Highest Booking Month	August

5. Tools and Analysis Workflow

The project follows the required end-to-end workflow: Python and Pandas are used for data loading, cleaning, transformation and basic analysis; BeautifulSoup is used to collect approximately 10 external tourism/informational records; PostgreSQL is used for the five business SQL questions; Excel is used for pivot analysis, formulas, conditional formatting and monthly trend reporting; and Power BI is used to build the one-page interactive management dashboard.

6. Dashboard Coverage

- KPI cards: Total Bookings, Successful Bookings, Cancellation Rate, Average ADR and Average Stay Duration.

- Monthly Booking Trend.
- Cancellation Rate by Hotel.
- Top 10 Countries by Bookings.
- Average ADR by Customer Type.
- Cancellation Rate by Lead Time.
- City Hotel vs Resort Hotel Bookings.
- Interactive slicers for Hotel, Month, Customer Type and Country.

7. Business Recommendations

- **1. Target long-lead bookings with stronger confirmation controls.** Use timely reminders, confirmation checkpoints and suitable deposit policies for higher-risk long-lead bookings.
- **2. Prioritize cancellation management for City Hotel.** Review channels, customer segments and booking policies contributing to the higher cancellation level before applying broad restrictions.
- **3. Protect high-ADR customer demand.** Use targeted packages, value-added offers and demand-based pricing to retain higher-value demand without unnecessary blanket discounts.

8. Conclusion

The analysis is designed to identify the main drivers of hotel cancellations and booking behaviour. The combined Python, BeautifulSoup, PostgreSQL, Excel and Power BI workflow provides a practical decision-support approach for hotel management, with particular attention to booking volume, hotel-level cancellation patterns, monthly demand and higher-value customer segments.

Prepared for Project 2 according to the requirements in the uploaded Data Analyst Project Assessment. The layout follows the uploaded Project Summary reference format.